

Assignment 1

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Problem 1

(a) Derive the result

$$\int d^4z \frac{\delta^2 W[J]}{\delta J(x) \delta J(z)} \frac{\delta^2 \Gamma[\Phi]}{\delta \Phi(z) \delta \Phi(y)} = -\delta^{(4)}(x - y), \quad (1)$$

and interpret diagrammatically. Here, $W[J]$ is the generating functional for the connected Green functions and $\Gamma[\Phi]$ is the generating functional for the one particle irreducible (1PI) Green functions.

(b) By taking one further functional derivative, show that Γ generates the amputated connected three-point function.

Solution

(a) We begin with the generic generating functional for n -point functions, $\mathcal{Z}$, defined for a given theory/Lagrangian as:

$$\mathcal{Z}[J] = \int \mathcal{D}\phi \exp \left[i \int d^4x (\mathcal{L}(\phi, \partial_\mu \phi, \dots) + J(x)\phi(x)) \right]. \quad (2)$$

We proceed to define $W[J] = i \log \mathcal{Z}[J]$ and from there, produce the Legendre transform of W , Γ :

$$\Gamma[\Phi] = W[J] - \int d^4x J(x)\Phi(x) \quad (3)$$

The " Φ " of interest is indeed $\Phi(x) = -\frac{\delta W[J]}{\delta J(x)}$, as one might expect of a Legendre transform (note that while it is in the form above, $W[J]$ is independent of it). Further, it is obvious upon inspection that $J(x) = \frac{\delta \Gamma[\Phi]}{\delta \Phi(x)}$. Given these relations, we may exploit the change of functions (or a functional change of variables) to prove the identity as follows:

$$\begin{aligned} \frac{\delta}{\delta J(x)} &= \int d^4z \frac{\delta \Phi(z)}{\delta J(x)} \frac{\delta}{\delta \Phi(z)} \\ &= - \int d^4z \frac{\delta^2 W[J]}{\delta J(x) \delta J(z)} \frac{\delta}{\delta \Phi(z)}. \end{aligned} \quad (4)$$

Now, we note that for a well behaved function f , $\frac{\delta f(y)}{\delta f(x)} = \delta^{(4)}(x - y)$, a fact that we proved in the second quarter of QFT. Thus, employing (4), with the assumption that J is well behaved produces

$$\begin{aligned} \frac{\delta J(y)}{\delta J(x)} &= - \int d^4z \frac{\delta^2 W[J]}{\delta J(x) \delta J(z)} \frac{\delta J(y)}{\delta \Phi(z)} \\ &= - \int d^4z \frac{\delta^2 W[J]}{\delta J(x) \delta J(z)} \frac{\delta^2 \Gamma[\Phi]}{\delta \Phi(z) \delta \Phi(y)}, \end{aligned} \quad (5)$$

Problem 2

Consider a quantum field theory of a real scalar field governed by the Lagrangian density,

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 - \frac{\lambda}{4!} \phi^4 \quad (11)$$

(a) Evaluate perturbatively the generating functional for the connected Green functions, $W[J]$, keeping all terms up to and including terms of $\mathcal{O}(\lambda)$ as follows. First, show that $\mathcal{Z}[J] \equiv \exp iW[J]$ can be written in the following form,

$$\mathcal{Z}[J] = \mathcal{N} \left[1 - \frac{i\lambda}{4!} \int d^4y \left(\frac{1}{i} \frac{\delta}{\delta J(y)} \right)^4 + \mathcal{O}(\lambda^2) \right] \exp -\frac{i}{2} \int d^4x_1 d^4x_2 J(x_1) \Delta_F(x_1 - x_2) J(x_2), \quad (12)$$

where $\mathcal{N}$ is the J -independent constant. Then, carry out the functional derivatives with respect to J , keeping all terms up to and including terms of $\mathcal{O}(\lambda)$. Using the result just obtained for $\mathcal{Z}[J]$, obtain an expression for $W[J]$ keeping all terms up to and including terms of $\mathcal{O}(\lambda)$.

(b) Using the result of part (a) for $W[J]$, compute the four-point connected Green function. By taking the appropriate Fourier transform, derive the momentum space Feynman rule for the four-point scalar interaction.

(c) Evaluate perturbatively the classical field $\phi_c(x)$ and the generating functional for the 1PI Green functions, $\Gamma[\phi_c]$, keeping all terms up to and including terms of $\mathcal{O}(\lambda)$. Then, repeat part (b) for the four-point 1PI Green function.

Solution

(a) We split the Lagrangian in the usual way, $\mathcal{L} = \mathcal{L}_0 + \mathcal{L}_I$, where $\mathcal{L}_I = -\frac{\lambda}{4!} \phi^4$. We then define the generating functional in the usual way (see (2)). Here, we can split the generating functional into the usual free field part, and an interacting part. Note that, using some rearrangement and integration by parts, we can write the free field generating functional¹ in terms of the Feynman propagator:

$$\begin{aligned} \mathcal{Z}_0[J] &= \mathcal{N} \exp \left(i \int d^4x \mathcal{L}_0 + J(x) \phi(x) \right) \\ &= \mathcal{N} \exp \left(-\frac{i}{2} \int J(x) \Delta_F(x - y) J(y) d^4x d^4y \right) \end{aligned} \quad (13)$$

To see the equivalence in (13), we note that we can switch $\partial_\mu \phi \partial^\mu \phi$ out with $-\phi \square \phi$ by integrating by parts, and the rest simply follows from completing the square and defining $\Delta_F(x - y) \equiv -(\square + m^2)^{-1}$ in the usual way².

Note that $\mathcal{L}_0$ is independent of $J(x)$. Thus, we can bring down ϕ^n by operating on $\mathcal{Z}_0[J]$ with the operator $(-i \frac{\delta}{\delta J})^n$. Since we can bring down as many powers of ϕ as necessary, we can create a Taylor series (or more generally, a power series) for any analytic function of ϕ , including $\mathcal{L}_I$. We may then use this idea to construct the full generating functional. Instead of ascertaining each coefficient in the Taylor series, we note that all of this information is already encoded in $\mathcal{L}_I$, allowing us to simply substitute the $(-i \frac{\delta}{\delta J})$ operator for ϕ in $\mathcal{L}_I$ to reconstruct it. Thus, we may write:

$$\mathcal{Z}[J] = \exp \left(i \int d^4x \mathcal{L}_I \left(-i \frac{\delta}{\delta J(x)} \right) \right) \mathcal{Z}_0[J] \quad (14)$$

Now comes the hard part: admitting that we don't know how to obtain an expression for (14). Luckily, we are physicists, and this doesn't bother us too much, for we still have a trick up our

¹In the second line of expression (13), there is often a $\det(-\square + m^2)^{-1/2}$ term in front of the exponential in the second line.

²It is actually $\Delta_F(x - y) = -(\square + m^2 - i\epsilon)^{-1}$, to ensure the invertibility of the operator, which is consistent with the $i\epsilon\phi^2$ term we add to ensure the convergence of the integral.

sleeves: Taylor expansion. We can Taylor expand the expression above in powers of the integral shown explicitly in (14) (assuming that our expansion parameter, the coupling in $\mathcal{L}_I$, is small enough). Clearly, this is 1 to zeroth order. Going to second order to show an explicit expression for this, we may write

$$\begin{aligned} \mathcal{Z}[J] = N \left[1 + \frac{1}{1!} i \int d^4x \mathcal{L}_I \left(-i \frac{\delta}{\delta J(x)} \right) \right. \\ \left. + \frac{1}{2!} i \int d^4y \mathcal{L}_I \left(-i \frac{\delta}{\delta J(y)} \right) i \int d^4z \mathcal{L}_I \left(-i \frac{\delta}{\delta J(z)} \right) + \dots \right] \mathcal{Z}_0[J] \end{aligned} \quad (15)$$

One may write the general expression for this expression in terms of a sum and a product within the sum, but it does not further our cause here. We are interested in the $\mathcal{O}(\lambda)$ results. Noting that each 'copy' of $\mathcal{L}_I$ contains one power of λ , it suffices to truncate this expression in our specific case to the first order, to obtain the desired result. Plugging in the exact expression for $\mathcal{L}_I$ gives us exactly the expression (12).

Now, we may explicitly take functional derivatives of the free generating functional to produce the first order results. Before we jump into the madness of 4 derivatives on a Gaussian, I will define the following:

$$F(x) = -\frac{i}{2} \left(\int d^4y J(y) \Delta_F(y-x) + \int d^4z \Delta_F(x-z) J(z) \right) \quad (16)$$

This is essentially the effect of taking the functional derivative of the argument of the exponential in (13) (with some indices relabeled). It is also useful to note that $\frac{\delta}{\delta J(x)} F(x)$ gives us $-i\Delta_F(0)$. Armed with these tools, we proceed onward.

Let us compute $\frac{\delta}{\delta J(x)} \mathcal{Z}_0[J]$:

$$\frac{\delta}{\delta J(y)} \mathcal{Z}_0[J] = -\frac{i}{2} \left(\int d^4x_1 J(x_1) \Delta_F(x_1 - y) + \int d^4x_2 \Delta_F(y - x_2) J(x_2) \right) \mathcal{Z}_0[J]$$

Note that this is simply $F(y) \mathcal{Z}_0[J]$. Taking another functional derivative, by the product rule gives us:

$$\left(\frac{\delta}{\delta J(y)} \right)^2 \mathcal{Z}_0[J] = F(y) F(y) \mathcal{Z}_0[J] + \left(\frac{\delta}{\delta J(y)} F(y) \right) \mathcal{Z}_0[J]$$

The second term is no mystery to us - it is simply $-i\Delta_F(0)$ as mentioned above. With this notation, it is easy to see³ that

$$\left(\frac{\delta}{\delta J(y)} \right)^4 \mathcal{Z}_0[J] = \mathcal{Z}_0[J] (F^4(y) - 6i\Delta_F(0)F^2(y) - 3\Delta_F^2(0)). \quad (17)$$

With this equation, we can finally provide an explicit form for $\mathcal{Z}[J]$ to $\mathcal{O}(\lambda)$. However, the item of interest here is the generating functional for the connected Green functions, which is (up to a factor of $-i$) the logarithm of $\mathcal{Z}[J]$. The logarithm separates the free field results from the perturbative interaction pieces. More explicitly,

$$\begin{aligned} iW[J] = -\frac{i}{2} \int J(x) \Delta_F(x-y) J(y) d^4x d^4y + \log N \\ + \log \left(1 - i \frac{i\lambda}{4!} \int d^4y (F^4(y) - 6i\Delta_F(0)F^2(y) - 3\Delta_F^2(0)) + \mathcal{O}(\lambda^2) \right). \end{aligned} \quad (18)$$

Note that $\log(1-x) \approx -x$ for $x \ll 1$, so assuming that $\lambda \ll 1$ allows us to use the first order approximation. Any higher order terms necessarily contain higher powers of λ . Similarly, there is

³If we define $F \equiv F(y)$ and $\Delta \equiv -i\Delta_F(0)$, the coefficient for the j^{th} power of F after n functional derivatives, $a_j^{(n)}$ follows the recursion relation $a_j^{(n)} = a_j^{(n-1)} + \Delta a_{j+1}^{(n-1)}$, with $a_j^{(1)} = \delta_{1,j}$

no need to worry about the higher order terms within the logarithm if we seek a first order result, since there is no reduction in the order while Taylor expanding.

The generating functional for connected Green functions is then given to first order by

$$W[J] = -i \log N + \frac{1}{2} \int J(x) \Delta_F(x-y) J(y) d^4x d^4y \quad (19)$$

$$- \frac{i\lambda}{4!} \int d^4y (F^4(y) - 6i\Delta_F(0)F^2(y) - 3\Delta_F^2(0)) + \mathcal{O}(\lambda^2).$$

(b) The connected Green functions are generated by $W[J]$, by operating on it with $-i \frac{\delta}{\delta J}$ and setting J to 0, implying that the connected 4-point is generated by 4 such functional derivatives. Now, to $\mathcal{O}(\lambda)$, we can determine the Green function by using (19). Note that $F(y)$ is linear in J . Taking 4 functional derivatives of any polynomial of J of order < 4 simply returns 0. Thus, we only have one surviving term.

$$G_c^{(4)}(x_1, x_2, x_3, x_4) = (-i)^4 \frac{\delta}{\delta J(x_4)} \frac{\delta}{\delta J(x_3)} \frac{\delta}{\delta J(x_2)} \frac{\delta}{\delta J(x_1)} \left(-\frac{i\lambda}{4!} \int d^4y F^4(y) \right) \quad (20)$$

One can easily work out that $\frac{\delta}{\delta J(x)} F^n(y) = -in\Delta_F(x-y)F^{n-1}(y)$, since Δ_F is an even function. This would imply then that

$$G_c^{(4)}(x_1, x_2, x_3, x_4) = -i\lambda \int d^4y \Delta_F(x_1-y) \Delta_F(x_2-y) \Delta_F(x_3-y) \Delta_F(x_4-y) \quad (21)$$

Now, note that the propagator satisfies $-(\square_x + m^2)\Delta_F(x-y) = i\delta^{(4)}(x-y)$, by definition⁴. Fourier transforming this relationship gives us that $(p^2 - m^2)\Delta_F(p) = i$. In other words, we have

$$\Delta_F(x-y) = \int \frac{d^4p}{(2\pi)^4} \frac{e^{ip(x-y)}}{p^2 - m^2 + i\epsilon} \quad (22)$$

Now, we can compute the Fourier transform of $G_c^{(4)}(x_1, x_2, x_3, x_4)$ with respect to each of the 4 variables. However, it is easier to plug in the explicit form for the propagators given above. This takes the form

$$\begin{aligned} \mathcal{F}[G_c^{(4)}] &= -i\lambda \int d^4x_1 d^4x_2 d^4x_3 d^4x_4 e^{i(\sum_i p_i x_i)} \int d^4y \int \prod_{n=1}^4 \frac{d^4k_n}{(2\pi)^4} \frac{e^{-ik_n(x_n-y)}}{k_n^2 - m^2} \\ &= -i\lambda \int d^4x_1 d^4x_2 d^4x_3 d^4x_4 e^{i(\sum_i p_i x_i)} \int \left(\prod_{n=1}^4 \frac{d^4k_n}{(2\pi)^4} \frac{e^{-ik_n x_n}}{k_n^2 - m^2} \right) \delta^{(4)}(\sum_n k_n) \\ &= -i\lambda \int \left(\prod_{n=1}^4 \frac{d^4k_n}{(2\pi)^4} \frac{i}{k_n^2 - m^2} \right) \delta^{(4)}(\sum_n k_n) \left(\prod_{i=1}^4 (2\pi)^4 \delta^{(4)}(p_i - k_i) \right) \\ &= -i\lambda \left(\prod_{n=1}^4 \frac{i}{p_n^2 - m^2} \right) \delta^{(4)}(\sum_{i=1}^4 p_i). \end{aligned}$$

This is our momentum space result. However, note that the δ function is present on the left as well for the momentum space Green function, which cancels out the one on the right. Note that ignoring the propagators and the conservation of 4-momentum gives us the Feynman rule in momentum space for a vertex: $-i\lambda$.

(c) The item of interest in this section is the classical field and the 1PI action, Γ . We need to find the classical field $\Phi(x) \equiv \frac{\delta}{\delta J(x)} W[J]$. We wish to obtain a result that is correct to $\mathcal{O}(\lambda)$, and the

⁴Again, with the usual $i\epsilon$ missing

terms excluded in our computation for $W[J]$ were all of $\mathcal{O}(\lambda^2)$. Since there is no order reducing procedure in the computation for Φ , the result (19) was all that was required to compute the first order Φ and Γ . Let us compute:

$$\frac{\delta}{\delta J(x)} W[J] \approx iF(x) - i\frac{\lambda}{4!} \int d^4y \left(-4i\Delta_F(x-y)F^3(y) - 12\Delta_F(0)\Delta_F(x-y)F(y) \right)$$

This implies that

$$\Phi(x) = iF(x) + i\frac{\lambda}{3!} \int d^4y \left(i\Delta_F(x-y)F^3(y) + 3\Delta_F(0)\Delta_F(x-y)F(y) \right) + \mathcal{O}(\lambda^2). \quad (23)$$

Now, we can Legendre transform $W[J]$ to get the generating functional for the $1PI$ diagrams $\Gamma[\Phi] = W[J] - \int d^4x J(x)\Phi(x)$. I will not write the explicit expression for this out, as all the elements have been written out to the desired order, and the expression is quite long. Now, we are interested in the 4-point $1PI$ interaction. This is given by operating on $\Gamma[\Phi]$ with $\frac{\delta}{\delta\Phi}$, 4 times. Now, since $W[J]$ and $\Gamma[\Phi]$ are Legendre transforms of each other, constructed with independent variables, $\frac{\delta}{\delta\Phi(x)} W[J] = 0$. This is an exact result.

We now need to invert equation (23) to get J in terms of Φ . To zeroth order, we have the relation $\Phi_x = (\square + m^2)_{xy}^{-1} J_y$, which implies that $J = (\square + m^2)\Phi$. Now, we can dial λ back up to a non-zero value with the relationship $(\square + m^2)\Phi(x) = J(x) + \mathcal{O}(\lambda)$, where the $\mathcal{O}(\lambda)$ terms are exactly as seen in (23). Now, in the $\mathcal{O}(\lambda)$ terms, we may plug in the solution $J(x) = (\square + m^2)\Phi(x)$, and integrate by parts to shift the $(\square + m^2)$ onto the Δ_F copies and get the answer in terms of $\Phi(x)$. This gives us:

$$J(x) = (\square + m^2)\Phi(x) + i\frac{\lambda}{2!}\Delta_F(0)\Phi(x) + \frac{\lambda}{3!}\Phi^3(x) + \mathcal{O}(\lambda^2) \quad (24)$$

Plugging this back into the Legendre transform⁵ formula for Γ , we get the result

$$\Gamma[\Phi] = -\frac{1}{2} \int d^4x \Phi(x) (\square + m^2) \Phi(x) - i\frac{\lambda}{4} \Delta_F(0) \int d^4x \Phi^2(x) - \frac{\lambda}{4!} \int d^4x \Phi^4(x) + \mathcal{O}(\lambda^2) \quad (25)$$

We may now take 4 functional derivatives of the effective action:

$$\frac{\delta}{\delta\Phi(x_4)} \frac{\delta}{\delta\Phi(x_3)} \frac{\delta}{\delta\Phi(x_2)} \frac{\delta}{\delta\Phi(x_1)} i\Gamma[\Phi] \Big|_{\Phi=0} = -i\lambda \int d^4x \left(\prod_{i=1}^4 \delta^{(4)}(x - x_i) \right) + \mathcal{O}(\lambda^2) \quad (26)$$

The Fourier transform is trivial in this case, and just returns a delta function of momentum conservation. Since our momentum space Green function already has a δ function of momentum conservation, which cancels the Fourier transform result:

$$i\Gamma^{(4)}(p) = -i\lambda \quad (27)$$

⁵This is also known as the Fenchel conjugate of $W[J]$, in mathematical literature, I believe.

Problem 3

Consider a quantum field theory of a real scalar field governed by the Lagrangian density,

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - V(\phi), \quad (28)$$

and the corresponding equation of motion,

$$\square \phi - V'(\phi) = 0,$$

where $\square \equiv \partial_\mu \partial^\mu$ and $V'(\phi) \equiv \frac{dV}{d\phi}$. The goal of this exercise is to derive the equation of motion for the Green function $\langle \Omega | T \{ \phi(x) \phi(y) \} | \Omega \rangle$,

$$\square_x \langle \Omega | T \{ \phi(x) \phi(y) \} | \Omega \rangle = - \langle \Omega | T \{ V'(\phi(x)) \phi(y) \} | \Omega \rangle - i \delta^{(4)}(x - y) \quad (29)$$

In order to obtain eq. (29), you should employ the following technique. Start from the path integral definition of the generating functional,

$$\mathcal{Z}[J] = N \int \mathcal{D}\phi \exp \left[i \int d^4x (\mathcal{L} + J(x) \phi(x)) \right], \quad (30)$$

where N is chosen such that $\mathcal{Z}[0] = 1$. Perform a change of variables in the path integral, $\phi(x) \rightarrow \phi(x) + \varepsilon(x)$, where $\varepsilon(x)$ is an arbitrary infinitesimal function of x . Noting that a change of variables⁶ does not change the value of $\mathcal{Z}[J]$, show that to first order in $\varepsilon(x)$,

$$\int \mathcal{D}\phi \exp \left(i \int d^4x [\mathcal{L} + J(x) \phi(x)] \right) \int d^4x \varepsilon(x) [-\square \phi - V'(\phi) + J(x)] = 0. \quad (31)$$

Since $\varepsilon(x)$ is arbitrary, we may choose $\varepsilon(x) = \epsilon \delta^{(4)}(x - y)$, where ϵ is an infinitesimal constant. With this choice for $\varepsilon(x)$, show that by taking the functional derivative of the eq. (31) with respect to $J(x)$ and then setting $J = 0$, one ends up with eq. (30).

HINT: What is the Jacobian corresponding to the change of variables, $\phi(x) \rightarrow \phi(x) + \varepsilon(x)$?

Solution

One could simply resort to the general Schwinger-Dyson equations to produce this equation along with the contact terms. However, we will be doing one better⁷, and proving a specific sub-case.

We know that the change of variables does not affect the actual value of the path integral itself, so we may make the substitution with no change in the quantity represented. Now, all we need to successfully change the variables out in the path-integral is the Jacobian term. Adding some arbitrary "fixed" function to ϕ is the equivalent of shifting a variable by a constant in a regular integral - the Jacobian is simply 1 (or the determinant of the identity). This can also be represented by the fact that $\frac{\delta \varepsilon}{\delta \phi} = 0$, implying that $\mathcal{D}(\phi + \varepsilon) = \mathcal{D}\phi$. Thus,

$$\mathcal{Z}[J] = N \int \mathcal{D}\phi \exp \left[i \int d^4x (\mathcal{L}(\phi + \varepsilon, \partial_\mu \phi + \partial_\mu \varepsilon) + J(x) \phi(x) + J(x) \varepsilon(x)) \right]. \quad (32)$$

Since this is equal to (30), we may extract the ε pieces from (32), and they should all equal 0, since ε is the only change made. However, this is easier said than done, given that we need to deal with an arbitrary function of $\phi + \varepsilon$. To this end, we assume that $V \in \mathcal{C}^\infty(\mathbb{C})$ to be able to expand out around ϕ and isolate it⁸. Further, we will consider $\varepsilon \ll \phi$ so that we may consider this treatment to

⁶Just as in the case of ordinary integration, a change of integration variables does not change the value of the functional integral.

⁷Subject to opinion.

⁸This might be a more strict condition than necessary.

first order alone, and I will work piece by piece to do so.

To expand to first order in ε , note that the $J(x)\varepsilon(x)$ piece separates out very easily. However, this is multiplied into the generating functional, and does not actually give us an additional term if simply "left alone". Thus, we may expand to first order to get $\exp \left[i \int d^4x J(x)\varepsilon(x) \right] \approx 1 + i \int d^4x J(x)\varepsilon(x)$.

It would be nice if we could extract a similar result for the "shifted" action term. To do so, note that it is sufficient to find a first order solution to this separation problem. Following the usual Euler-Lagrange mechanism, we use

$$\int d^4x \mathcal{L}(\phi + \varepsilon, \partial_\mu \phi + \partial_\mu \varepsilon) = \int d^4x \left(\mathcal{L}(\phi, \partial_\mu \phi) + \frac{\partial \mathcal{L}(\phi, \partial_\mu \phi)}{\partial \phi} \varepsilon + \frac{\partial \mathcal{L}(\phi, \partial_\mu \phi)}{\partial \partial_\mu \phi} \partial_\mu \varepsilon + \mathcal{O}(\varepsilon^2) \right).$$

Since we don't wish to deal with a derivative of ε , we use integration by parts, assuming that all boundary terms vanish. This gives us the usual $-\partial_\mu$ acting on $\frac{\partial \mathcal{L}}{\partial \partial_\mu \phi}$. Now, noting that we have more terms that look like $\exp \left[i \int d^4x \mathcal{J}(x)\varepsilon(x) \right]$, we may once again put it in the form of $1 + i \int d^4x \mathcal{J}(x)\varepsilon(x)$ to first order in ε . Identifying $\mathcal{J} = \frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial \partial_\mu \phi}$, we may write

$$\mathcal{Z}[J] = N \int \mathcal{D}\phi \exp \left[i \int d^4x (\mathcal{L} + J\phi) \right] \left(1 + i \int d^4x \mathcal{J}(x)\varepsilon(x) \right) \left(1 + i \int d^4x J(x)\varepsilon(x) \right) + \mathcal{O}(\varepsilon^2) \quad (33)$$

where it must be noted that there is still a term that is $\mathcal{O}(\varepsilon^2)$ that has been written explicitly, and must be tossed out. To provide a more familiar form, we need to explicitly find $\mathcal{J}$. Given the form of the Lagrangian density, $-\partial_\mu \frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} = -\square \phi$, and $\frac{\partial \mathcal{L}}{\partial \phi} = -V'(\phi)$ implying $\mathcal{J} = -(\square + V'(\phi))$. Now, we get the result

$$\mathcal{Z}[J] = \int \mathcal{D}\phi \exp \left[i \int d^4x (\mathcal{L} + J(x)\phi(x)) \right] \left(1 + i \int d^4x \varepsilon(x) [-\square \phi - V'(\phi) + J(x)] + \mathcal{O}(\varepsilon^2) \right) \quad (34)$$

which directly implies (31) as the $\mathcal{O}(\varepsilon)$ term has to vanish for the equality to hold.

Now, we can choose a form for the arbitrary ε . As prescribed, we pick $\varepsilon(x) = \delta^{(4)}(x - y)$ to further the cause of this expansion. Now, we proceed to take a functional derivative of eq. (31) with respect to J , i.e. we operate on it with $\frac{\delta}{\delta J(z)}$. Employing the product and chain rules for functionals, we get

$$0 = N \int \mathcal{D}\phi \exp \left[i \int d^4x (\mathcal{L} + J(x)\phi(x)) \right] \left(i \int d^4x \phi(x) \delta^{(4)}(x - z) \int d^4w \varepsilon(w) [\mathcal{J}(x) + J(x)] + \int d^4x \varepsilon(x) \delta^{(4)}(x - z) \right)$$

Now, we plug in the functional form of ε in order to retrieve the result

$$0 = N \int \mathcal{D}\phi \exp \left[i \int d^4x (\mathcal{L} + J\phi) \right] \left(i \epsilon \left[-\phi(z) \square_y \phi(y) - \phi(z) V'(\phi(y)) + \phi(z) J(y) - i \delta^{(4)}(y - z) \right] \right)$$

Setting $J = 0$ and relabeling some indices gives us

$$0 = \int \mathcal{D}\phi e^{iS[\phi, \partial_\mu \phi]} \left(\square_x \phi(x) \phi(y) + V'(\phi(x)) \phi(y) + i \delta^{(4)}(x - y) \right). \quad (35)$$

This can be rewritten in the desired notation.

$$\square_x \langle \Omega | \phi(x) \phi(y) | \Omega \rangle = - \langle \Omega | V'(\phi(x)) \phi(y) | \Omega \rangle - i \delta^{(4)}(x - y) \quad (36)$$

Problem 4

Consider a theory of a real scalar field governed by eq. (28) with $V(\phi) = \frac{1}{2}m^2\phi^2$.

(a) Compute exactly the free-field Feynman propagator, $\Delta_F(x)$, in coordinate space.

HINT: You will need to consult a good table of integrals such as I.S. Gradshteyn & I.M. Ryzhik, Table of Integrals, Series and Products. You should be able to find integrals that when differentiated with respect to a parameter yield the integral of interest for this problem.

(b) Evaluate the leading singularities of $\Delta_F(x)$ near the light cone, $x^2 = 0$.

(c) Using the Källén–Lehmann representation, comment on the leading singularity of the exact two-point function near the light cone.

Solution

(a) Integrating the Lagrangian density (28) by parts gives us the action:

$$S = -\frac{1}{2} \int d^4x \phi(x)(\square + m^2)\phi(x) \quad (37)$$

Noting that the effective action is just the action (up to a renormalized mass correction) at tree-level, we may immediately skip to the defining equation for the propagator by taking functional derivatives, and defining a Green's function for the inversion of the result:

$$-(\square + m^2)\Delta_F(x - y) = \delta^{(4)}(x - y)$$

We may Fourier transform this equation with respect to x . This produces

$$(p^2 - m^2)\Delta_F(p) = 1 \quad (38)$$

In turn, this produces the solution

$$\Delta_F(x) = \int \frac{d^4p}{(2\pi)^4} \frac{e^{-ipx}}{p^2 - m^2} \quad (39)$$

where this is simply the Fourier Transform of $\frac{1}{p^2 - m^2}$. We look at the table of integrals by Gradshteyn, Ryzhik and Zwillinger, to identify Fourier pairs, $(f(x), \mathcal{F}[f](\xi))$ to apply to the integral. To use this, we must work without the metric trickery. Rewriting the integral above gives:

$$\Delta_F(x) = \int \frac{d^3p}{(2\pi)^3} e^{i\vec{p}\vec{x}} \int \frac{dp_0}{2\pi} \frac{e^{-ip_0x_0}}{p_0^2 - (\vec{p}^2 + m^2)}$$

Note that $\vec{p}^2 + m^2 \geq 0$ by positive definiteness of the Euclidean sub-metric and the fact that $m \in \mathbb{R}$. It is physically realised as the fact that $E_p^2 \geq 0$. The p_0 integral is simple, and can be integrated about its poles with the usual contour methods, by sending $E_p^2 \rightarrow E_p^2 - i\epsilon$ (consistent with the usual Feynman picture) and integrating around the pole, depending on the sign of x_0 . This gives us a residual value when evaluated. However, being lazy, I prefer using the table for Fourier pairs⁹. For this, we still need to use the $i\epsilon$. This integral gives us the result (after taking $\epsilon \rightarrow 0$):

$$\Delta_F(x) = \int \frac{d^3p}{(2\pi)^3} e^{i\vec{p}\vec{x}} \frac{e^{iE_p|x_0|}}{2iE_p}$$

We may convert this to an integral in spherical coordinates, moving to the radial coordinate $P \equiv |\vec{p}|$.

$$\Delta_F(x) = \left(\int_0^{2\pi} \frac{d\varphi}{2\pi} \right) \int_0^\infty \frac{P^2 dP}{(2\pi)^2} \frac{e^{-i|x_0|\sqrt{P^2+m^2}}}{2i\sqrt{P^2+m^2}} \int_{-1}^1 d\cos\theta e^{iP|\vec{x}|\cos\theta}$$

⁹Page 1089, Table 12.23, Index 15.

Two of these are trivial integrals, and can be processed easily.

$$\Delta_F(x) = \int_0^\infty \frac{dP}{(2\pi)^2} P^2 \frac{e^{-i|x_0|\sqrt{P^2+m^2}}}{i\sqrt{P^2+m^2}} \frac{\sin(P|\vec{x}|)}{P|\vec{x}|} \quad (40)$$

This can be put into a convergent form at large P , since it currently goes like *const · oscillations* for $P \gg m$.

$$\Delta_F(x) = -\frac{1}{i(2\pi)^2|\vec{x}|} \partial_b \int_0^\infty dP \frac{e^{-\beta\sqrt{\gamma^2+P^2}}}{\sqrt{\gamma^2+P^2}} \cos bP \quad (41)$$

where $\beta = i|x_0|$, $\gamma = m$, and $b = |\vec{x}|$. The integral in (41) now goes like $\frac{1}{P} \cdot \text{oscillations}$ for $P \gg m$. Comparing this with the result in Gradshteyn, Ryzhik and Zwillinger's table¹⁰, we identify the solution to the integral presented. However, there is a slight issue with this: the integral table requires $\text{Re}(\beta), \text{Re}(\gamma), b > 0$. γ and b cause no issues, since they meet the criteria, but β is entirely imaginary. As a result, we must introduce a small parameter, $\delta > 0$, to ensure convergence, and retrieve the solution in the limit $\delta \rightarrow 0$. Thus, we set $\beta = \delta + i|x_0|$, ensuring convergence.

$$\Delta_F(x) = \lim_{\delta \rightarrow 0} -\frac{1}{i(2\pi)^2|\vec{x}|} \frac{\partial}{\partial b} K_0\left(\gamma\sqrt{\beta^2+b^2}\right) \quad (42)$$

where K_0 is a modified Bessel function of the second kind. However, we now have to understand our solution in different cases - space-like and time-like. This becomes more obvious when one notes that $\beta^2 + b^2$ is simply $-x_\mu x^\mu$, and we must break into the case of whether the quantity is positive or negative, within the square root. In the simpler case of a space-like separation, we are dealing primarily with positive, well behaved quantities, and we may send δ to 0, and take the derivative as usual. This gives us the propagator for space-like separation:

$$\Delta_F(x) = \frac{m}{i4\pi^2\sqrt{-x^2}} K_1\left(m\sqrt{-x^2}\right) \quad (43)$$

where $x^2 \equiv x_\mu x^\mu$ as usual¹¹.

While dealing with the case of $x^2 > 0$, we employ a bit of care, and note that $\beta^2 + b^2 < 0$, a cause for consternation¹². In this case, note that the modified Bessels of the second kind can be defined in terms of the Hankel functions¹³:

$$K_\alpha(z) = \begin{cases} \frac{\pi}{2} i^{\alpha+1} H_\alpha^{(1)}(iz) & -\pi < \arg(z) \leq \frac{\pi}{2} \\ \frac{\pi}{2} (-i)^{\alpha+1} H_\alpha^{(2)}(iz) & -\frac{\pi}{2} < \arg(z) \leq \pi \end{cases} \quad (44)$$

There is a bit of a nuance on which branch cut we are interested in taking. In fact, this is decided by the $i\epsilon$ we left out. Plugging it in and noting that $\epsilon > 0$ allows us to find the argument of the function $\sqrt{-x^2 + i\epsilon}$ (since m is real, it does not add to the argument). Note that eq.(44) has argument z , allowing us to identify $z = m\sqrt{-x^2 + i\epsilon}$. Since this is the case of $x^2 > 0$, we note that the argument inside the square root is $\arctan(-\frac{\epsilon}{x^2})$. Since ϵ is infinitesimally small, we may use the fact that $\arctan(y) = y + \mathcal{O}(y^3)$. Thus, the argument is some small negative angle. The square root is just a factor of $\frac{1}{2}$. This allows us to pick out a working conversion. Note that $\arg(z) \in (-\frac{\pi}{2}, 0]$, where both expressions are equally valid. Since the Hankel function of the second kind has analytic continuity into the space-like regime, we pick $H^{(2)}$. With this, we may finally provide the Feynman propagator in position space:

$$\Delta_F(x) = \Theta(-x^2) \frac{m}{i4\pi^2\sqrt{-x^2}} K_1\left(m\sqrt{-x^2}\right) + \Theta(x^2) \frac{m}{8\pi\sqrt{x^2}} H_1^{(2)}\left(m\sqrt{x^2}\right) \quad (45)$$

where Θ is the usual Heaviside step function.

¹⁰Page 495, Table 3.914, Index 4.

¹¹If one kept the $i\epsilon$ in, this would give $-x^2 + i\epsilon$, a result one could achieve by simply Wick rotating in the usual way: $x^2 \rightarrow x^2 - i\epsilon$ and $p^2 \rightarrow p^2 + i\epsilon$

¹²Or not. Subject to opinion. I just wanted to employ some alliteration for fun.

¹³I stole this expression from Wikipedia. Verified in G&S.

(b) I will try to keep the analysis contained to the expressions we have seen here, and then compare it with the result of expanding the functions. We go back to the last point that we see any $x_\mu x^\mu$, or a square root of it in the integral, eq. (40). Note that expression (40) can be rewritten in a slightly different way, by swapping the cos out for exponentials:

$$\Delta_F(x) = -\frac{1}{i4\pi^2|\vec{x}|} \int_0^\infty dP \frac{P}{2E_p} \left[e^{-i(E_p|x_0|-P|\vec{x}|)} - e^{-i(E_p|x_0|+P|\vec{x}|)} \right] \quad (46)$$

Observe that this integral is bounded (in absolute value) above by the integral of $\frac{P}{\sqrt{P^2+m^2}}$ for any upper limit. For small values of P , this clearly does not contribute much, as $m \gg P$. It is only at large 3-momenta that the bounding integral diverges linearly. This implies that this is the regime of interest. Thus, to study the leading order divergent behaviour, it is acceptable to consider the massless limit, $E_p \rightarrow P$. Our expression then simplifies quite a bit:

$$\Delta_F^{massless}(x) = -\frac{1}{i8\pi^2|\vec{x}|} \int_0^\infty dP \left(e^{-iP(|x_0|+|\vec{x}|)} - e^{-iP(|x_0|-|\vec{x}|)} \right) \quad (47)$$

Given that this is a purely oscillatory integral, one may think of this as "half of all the oscillations", resulting in a complete cancellation, unless either of $|x_0| \pm |\vec{x}|$ is 0. In other words, this is "half a δ -function". Given this, we write:

$$\lim_{|x_0| \rightarrow |\vec{x}|} \Delta_F(x) \approx -\frac{1}{8\pi^2 i |\vec{x}|} \cdot \frac{1}{2} (2\pi \delta(|x_0| - |\vec{x}|) - 2\pi \delta(|x_0| + |\vec{x}|))$$

Had there not been a - sign, we would have been able to combine the δ -functions into $|\vec{x}| \delta(x^2)$. However, this is one result for the leading singularity in the light cone. However, if approximating half a δ -function made one uncomfortable, they might consider instead, that eq. (47) is an exact expression for a Laplace transform from P space to $i(|x_0| \pm |\vec{x}|)$ space, for the unit element. However, for the transform to exist, we need $\text{Re}(i(|x_0| \pm |\vec{x}|)) > 0$. So in the usual fashion, we send $|x_0| \rightarrow |x_0| - i\xi$ and work in the limit of $\xi \rightarrow 0$. The Laplace transform of 1 into s space is simply $\frac{1}{s}$. Working in the aforementioned limit then gives us a simplification of eq. (47). Combining the $(i(|x_0| + |\vec{x}|))^{-1} + (i(|x_0| - |\vec{x}|))^{-1}$ and cancelling out the factors of i gives us:

$$\lim_{|x_0| \rightarrow |\vec{x}|} \Delta_F(x) \approx \frac{1}{4\pi^2} \frac{1}{x^2} \quad (48)$$

where $|x_0|^2 - |\vec{x}|^2 = x_\mu x^\mu = x^2$. This expression is much cleaner, and only suffers from the massless approximation, and working in the divergent limit (the latter is a necessity).

Finally, one may simply look at expression (47), and note that it can be converted to an expression for the Fourier transform of the Heaviside Theta function¹⁴ used earlier. This will allow us to formulate an exact expression for the massless limit and compare it with the approximate methods we have used above:

$$\lim_{|x_0| \rightarrow |\vec{x}|} \Delta_F(x) \approx \frac{1}{4\pi^2} \frac{1}{x^2} - \frac{1}{8\pi^2 i |\vec{x}|} \cdot \frac{1}{2} (2\pi \delta(|x_0| - |\vec{x}|) - 2\pi \delta(|x_0| + |\vec{x}|)) \quad (49)$$

Clearly, this exact expression has incorporated both of the approximations above, justifying, in part, each of them. Note that part of this singularity is due to the massless limit being taken.

Now, we may attempt to explore the singularities in the Bessel and Hankel functions as we approach the light-cone. Let us proceed with the analysis. We consider approaching the limit from the space-like case. Set $x^2 = -\frac{\zeta^2}{m^2}$ for some small $\zeta \in \mathbb{R}$, with $\zeta \ll m$. We look for an asymptotic form for

¹⁴Page 1088, Table 12.23, Index 6.

$K_1(\zeta)$ for small arguments. We produce¹⁵ $K_\nu(z) \sim \frac{1}{2}\Gamma(\nu)(\frac{1}{2}z)^{-\nu}$ for $\text{Re}(\nu) > 0$. Clearly, this implies that

$$\lim_{|x_0| \nearrow |\vec{x}|^-} \Delta_F(x) \approx \frac{1}{4\pi^2} \frac{1}{x^2}. \quad (50)$$

We move to the limit of the Hankel function (note that we could simply have used this expansion alone). Set $x^2 = \frac{\zeta^2}{m^2}$ with the same requirements on ζ as before. We now look at the expansion for $H_1^{(2)}(\zeta)$ as $\zeta \rightarrow 0$. We procure¹⁶ $iH_\nu^{(2)}(z) \sim -\frac{1}{\pi}\Gamma(\nu)(\frac{1}{2}z)^{-\nu}$ for $\text{Re}(\nu) > 0$. We can use this form with $\zeta = m\sqrt{x^2}$.

$$\lim_{|x_0| \searrow |\vec{x}|^+} \Delta_F(x) \approx \frac{1}{4\pi^2} \frac{1}{x^2} \quad (51)$$

Clearly, we have an agreement between both sides of the limits, and matches part of the limit we arrived at in the massless approximation. One might then attribute the δ -function light cone singularities to the massless limit that was taken. However, this should not be ignored. Light-like propagation is a property of massless particles (for the on-shell case at least), which implies that a δ -function should be included in the expression for the propagator. In fact, it is good to note that the singular expressions are independent of mass, which is in line with the preceding statement.

(c) We may simply plug in the result for the propagator into the expression for the Källén-Lehmann representation of the exact 2-point:

$$\langle 0 | \mathcal{T} \{ \phi(x) \phi(y) \} | 0 \rangle = \int_0^\infty d\mu^2 \rho(\mu^2) i \Delta_F(x - y; \mu^2) \quad (52)$$

where $\mathcal{T}$ is the usual time ordering operator in field-space, and ρ is the spectral density, given by

$$\rho(\mu^2) = \sum_\alpha \delta(\mu^2 - m_\alpha^2) | \langle 0 | \phi(0) | \alpha \rangle |^2 = Z_\phi \delta(\mu^2 - m_R^2) + \sigma(\mu^2) \quad (53)$$

where $Z_\phi = | \langle 0 | \phi(0) | 1 \text{ particle state} \rangle |^2$ is the usual wavefunction renormalization, $\sigma(\mu^2)$ is the spectral density of multi-particle (bound etc.) states. We may plug in the $\frac{1}{x^2}$ term, and obtain the result

$$\langle 0 | \mathcal{T} \{ \phi(x) \phi(y) \} | 0 \rangle \approx \frac{i}{4\pi^2(x-y)^2} \sum_{m_\alpha \in \mathbb{R}} | \langle 0 | \phi(0) | \alpha \rangle |^2 = \frac{i}{4\pi^2(x-y)^2} \left(Z_\phi + \int_0^\infty d\mu^2 \sigma(\mu^2) \right) \quad (54)$$

for small separations $(x-y)^2 \ll 1$. In the low energy limit, this simply becomes proportional to $\frac{iZ_\phi}{(x-y)^2}$, whereas the high energy limit sees greater contributions from the multi-particle spectral densities. Note that the δ -functions in the singularity are not going to be integrated, and will provide a similar result, but give a real contribution to the 2-point on the light-cone.

¹⁵Handbook of Mathematical Functions: with Formulas, Graphs, and Mathematical Tables by Milton Abramowitz and Irene Stegun, Page 375, 9.6.9.

¹⁶Abramowitz & Stegun, Page 360, 9.1.9.